

Standardizing Specialty Pharmacy Practices: Addressing Risk in Loading Dose and Maintenance Dose Management

Shelby Braginetz, PharmD, CSP | Marie McCreight, PharmD, CSP | Leslie DuVal, PharmD
University of Virginia Specialty & Home Delivery Pharmacy



Background

In a health system specialty pharmacy, recurring errors have been identified for loading and maintenance dose (LD/MD) prescriptions

A 39-year old patient with ulcerative colitis was prescribed a JAK-inhibitor for an 8-week loading dose, followed by a maintenance dose

Due to a lack of LD/MD standard operating procedures

Two refill requests were accepted on the LD script

Patient received 12 weeks of LD

No MD prescription was on file

Result: Excess high-dose JAK-inhibitor exposure, risk of adverse events, and delay in maintenance phase

This prompted an internal review of the possibility and occurrence of similar LD/MD errors across specialty pharmacy operations

High-Risk Workflow Vulnerability

- Inconsistent clinic prescribing practices
- No standardized documenting or tracking of LD/MD regimens
- Lack of clarity across intake, clinical review, and prescription processing

Clinical Significance

- Proper sequencing of LD/MD is essential for achieving clinical goals and maximizing cost-effectiveness
- Delays or misalignments can negatively impact quality-adjusted life years (QALYs), drive unnecessary healthcare resource utilization (HCRU), and compromise patient-reported outcomes (PROs)
- These risks can erode the long-term value of specialty therapies

Objectives

Primary ≥75% reduction in LD/MD-related near misses/errors

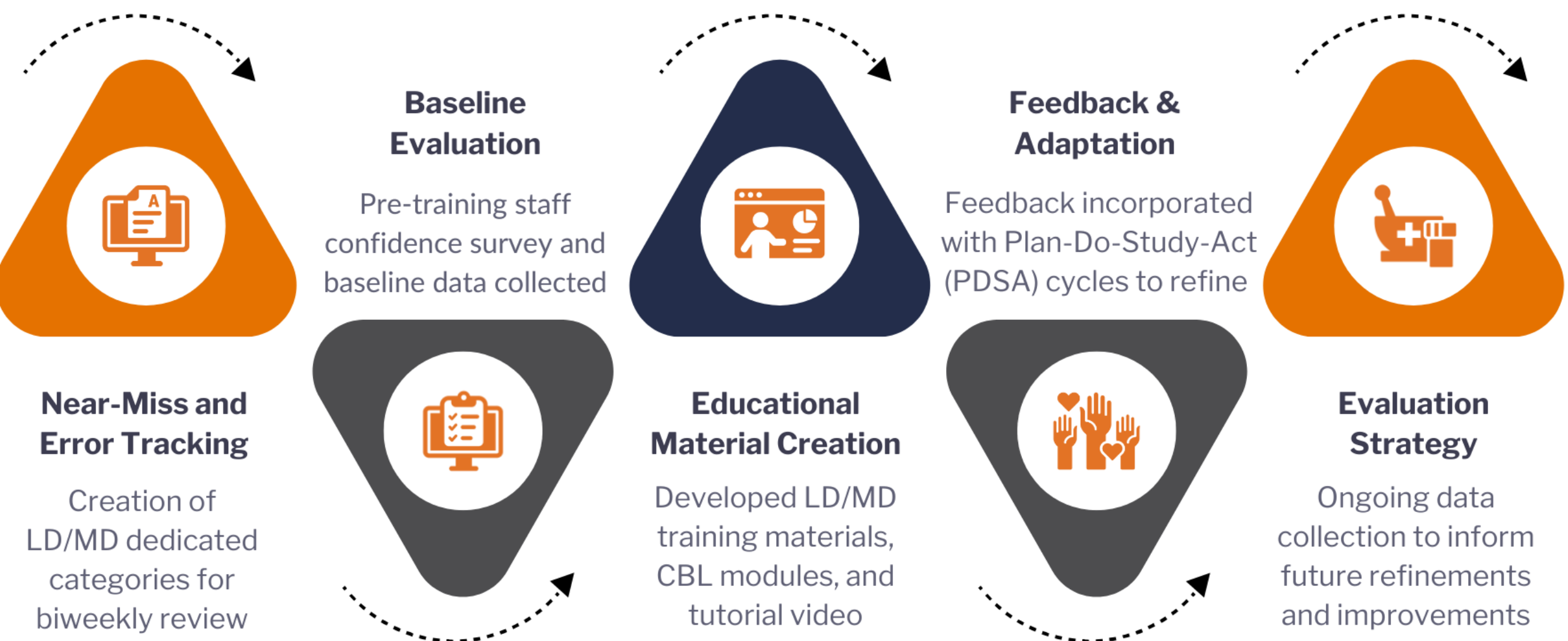
Secondary

- ≥25% increase in staff-reported confidence in managing complex dosing regimens
- ≥90% staff education and attestation completion rate within one month of launch

Tertiary

- LD/MD near-miss and error trend analysis
- Feasibility of proposed queue standardization
- Develop a long-term specialty pharmacy onboarding module for new hire pharmacists and technicians

Methodology

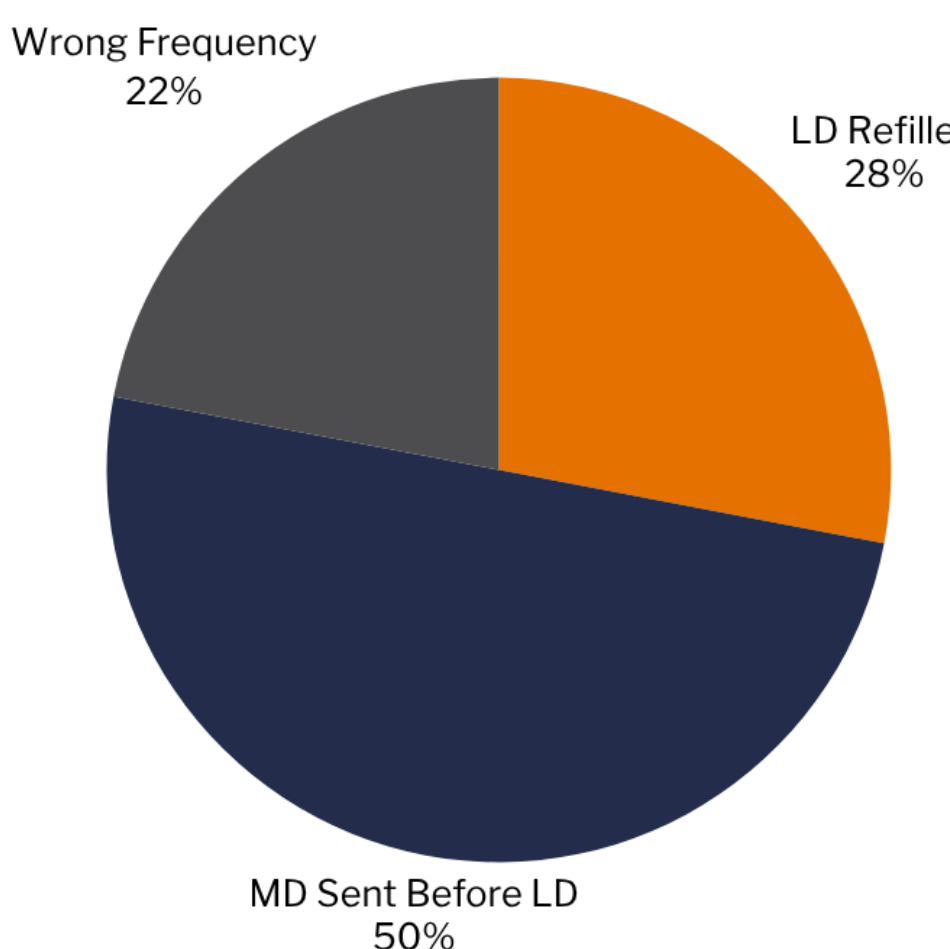


Future Considerations

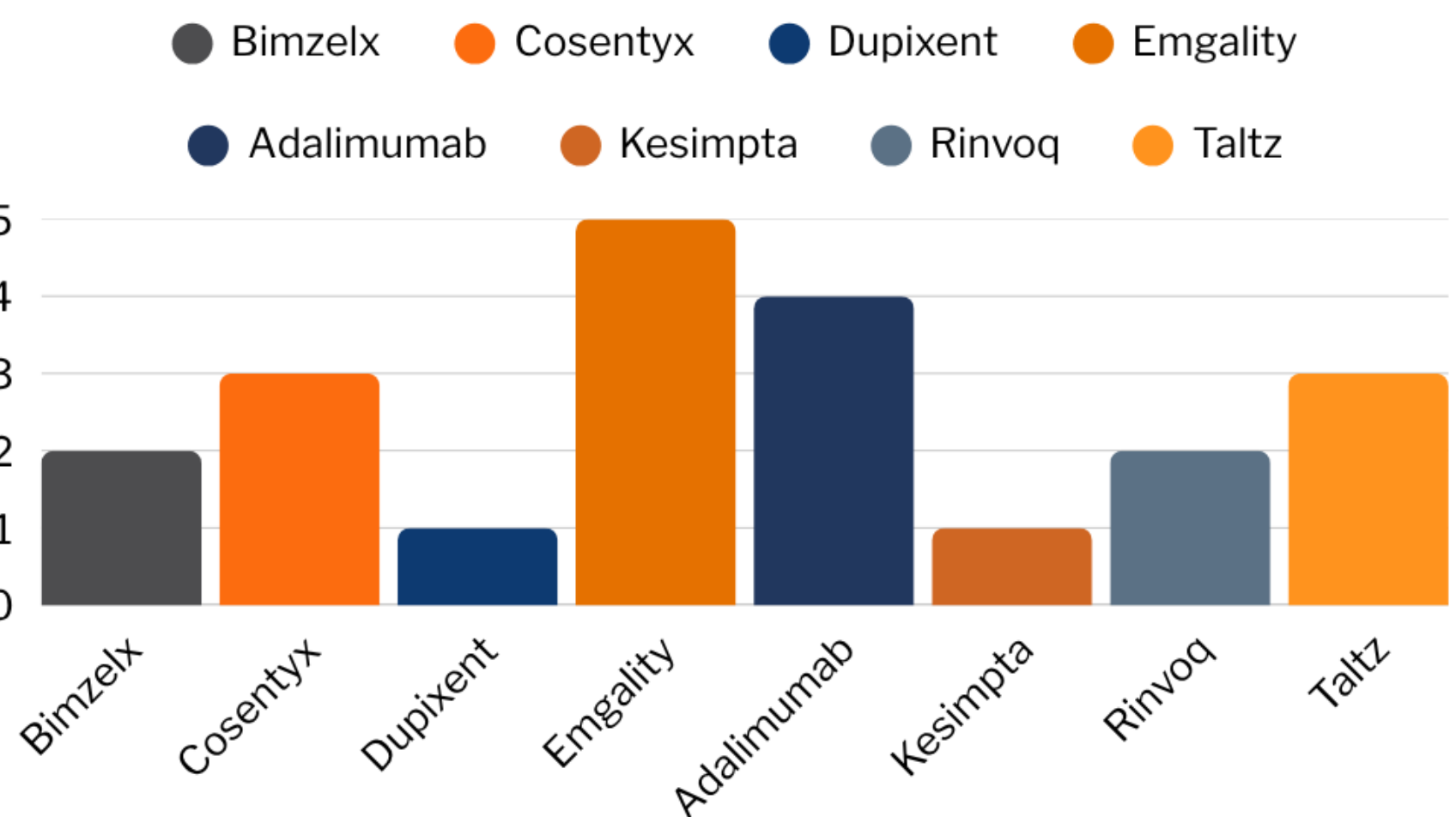
- Intervention not adopted widely** → Embed LD/MD logic into electronic medical record
- Errors persist despite training** → Restructure high-risk queues in Willow Ambulatory
- High refill error rate** → Limit autofill eligibility for loading doses

Baseline Data

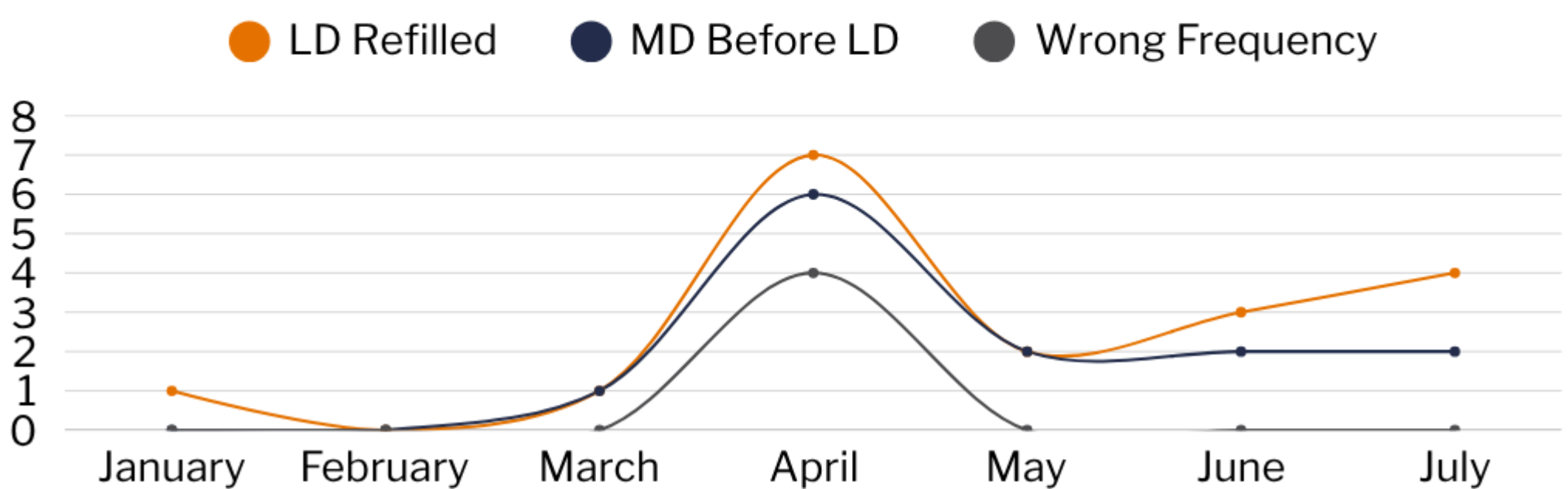
Types of LD/MD Errors



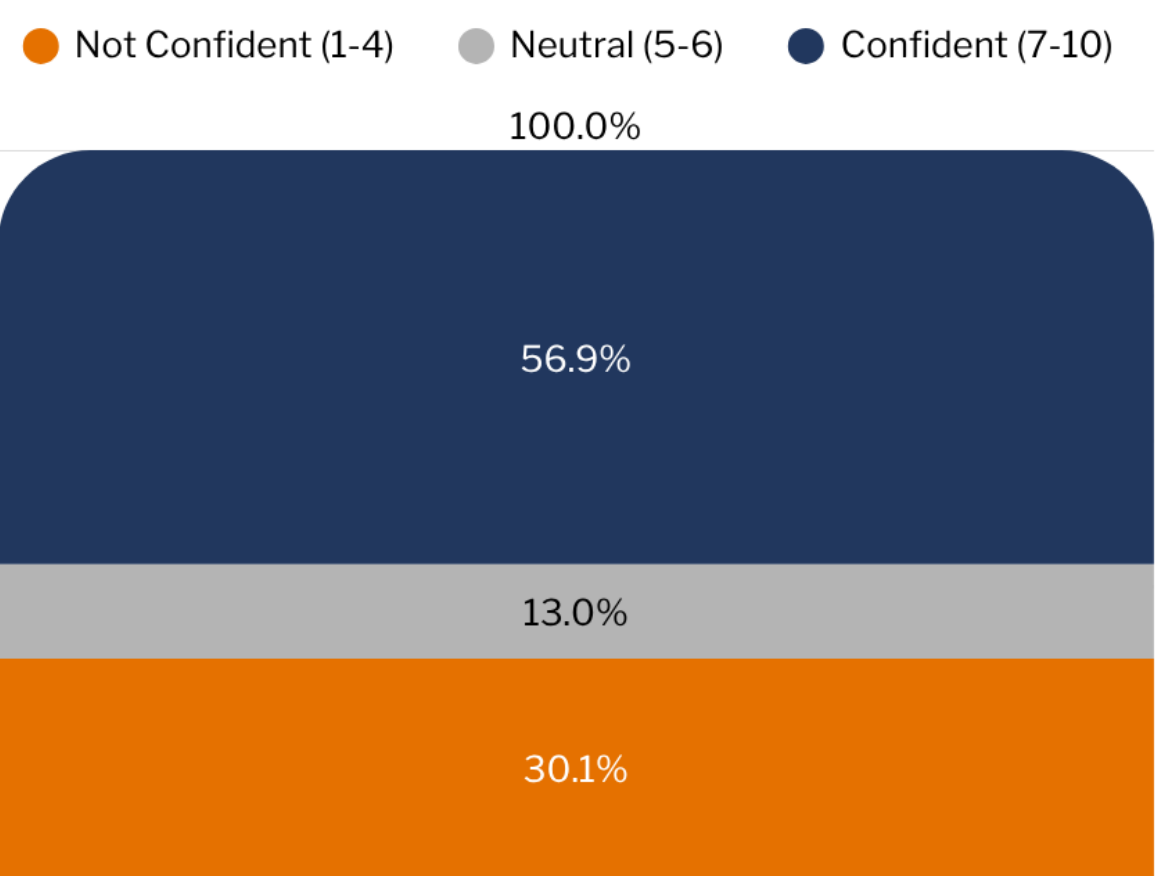
Medication Specific Near-Misses and Errors



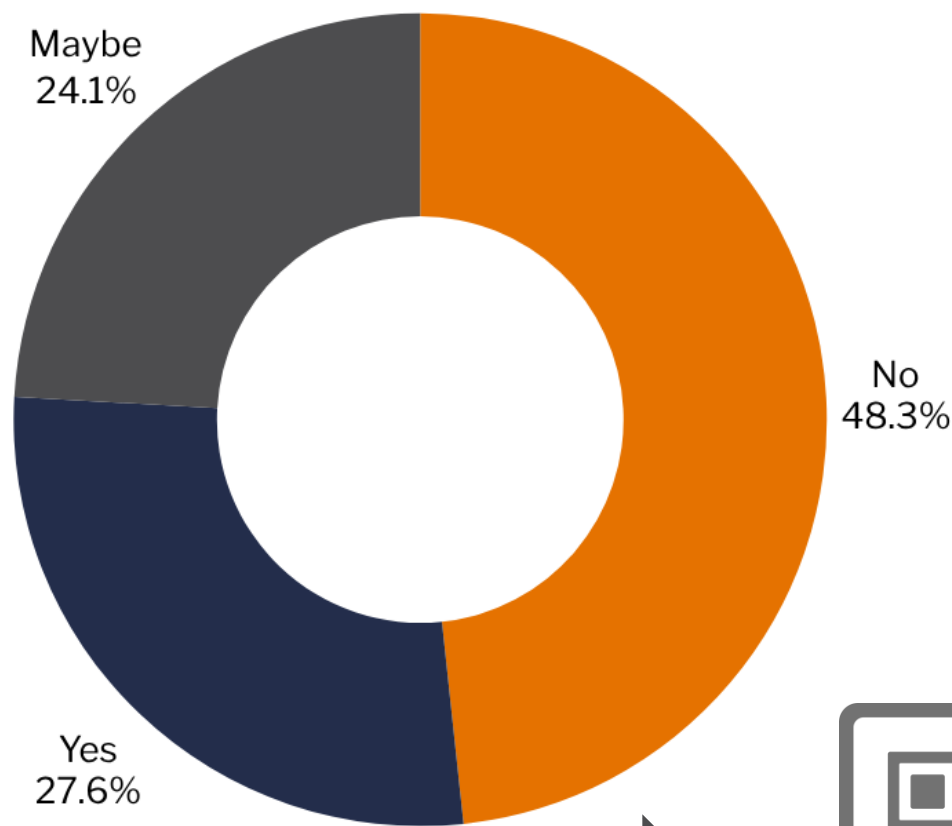
Trends in LD/MD Errors Over Time



Staff Confidence in LD/MD Workflow



Formal Training Exposure Rates



See the Training Document – Scan for Access

